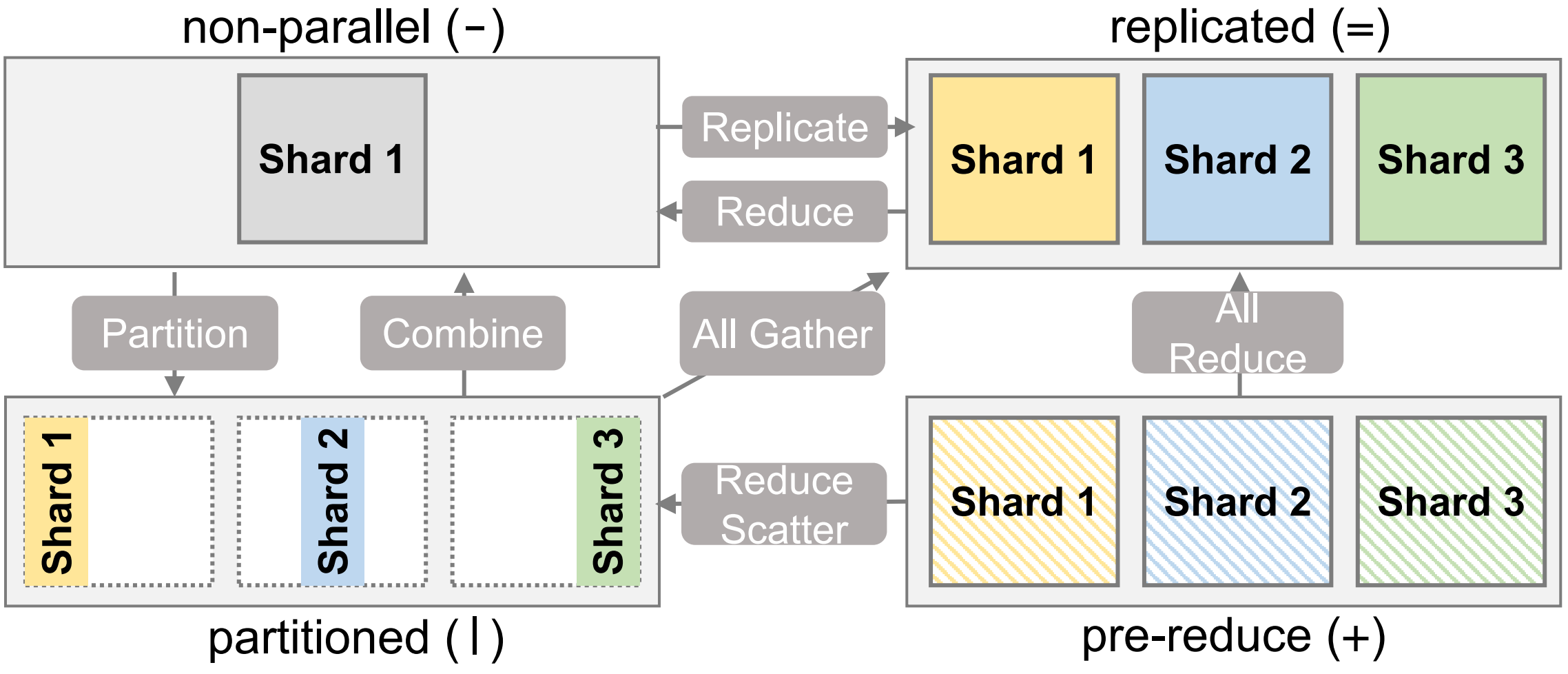
non-parallel (-)
replicated (=)
Shard 1
Replicate
Reduce
Shard 1
Shard 2
Shard 3
Partition
Combine
All Gather
All Reduce
Shard 1
Shard 2
Shard 3
Reduce Scatter
Shard 1
Shard 2
Shard 3
partitioned (|)
pre-reduce (+)